

FINA2204 Tutorial 4: Pricing futures contracts

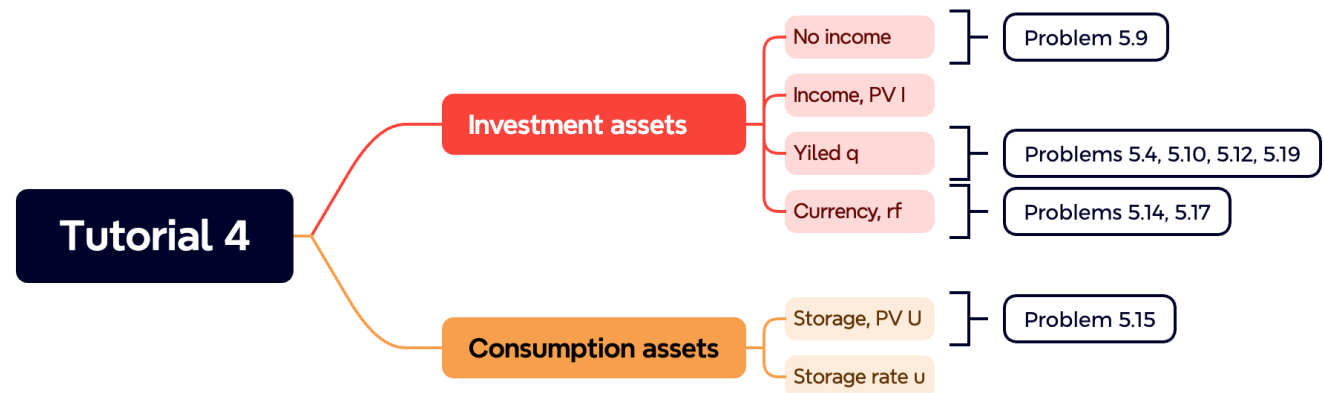
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Agenda

- Problem 5.4
- Problem 5.9
- Problem 5.10
- Problem 5.12
- Problem 5.14
- Problem 5.15
- Problem 5.17
- Problem 5.19



Presented with xmind

Recap: time value

Discrete compounding

- For an annual rate R_m , compounded m times a year, the future value of A after T years is:

$$A\left(1 + \frac{R_m}{m}\right)^{mT}$$

The power of frequency (m)

- *Annual* ($m = 1$) $\rightarrow$ *Semiannual* ($m = 2$) $\rightarrow$ *Monthly* ($m = 12$) $\rightarrow$ *Daily* ($m = 365$).
- As m increases, the future value grows because you earn “interest on interest” more frequently.

Recap: time value

Continuous compounding

- What happens if we compound every hour, every minute, every second... infinitely often?
- We let $m \rightarrow \infty$, and the formula simplifies to the continuous compounding formula:

$$Ae^{R_c T}$$

where R_c is the continuously compounded rate, and $e \approx 2.718$.

Recap: time value

Deriving $e^{R_c T}$

$$\text{goal} \rightarrow \lim_{m \rightarrow \infty} \left(1 + \frac{R_c}{m}\right)^{mT}$$

- Let $n = \frac{m}{R_c}$, as $m \rightarrow \infty$, $n \rightarrow \infty$.

$$\lim_{m \rightarrow \infty} \left[\left(1 + \frac{1}{n}\right)^n \right]^{R_c T}$$

- Since $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$

$$e^{R_c T}$$

Continuous compounding is the **limiting case** of increasingly frequent discrete compounding.

Recap: COC model

- We use the continuous compounding multiplier (e^{cT}) to connect the spot market with the forward or futures market.
- The relationship between futures prices (F_0) and spot prices (S_0) can be summarized in terms of **the cost of carry (COC)**.

$$F_0 = S_0 e^{cT} \quad (5.18)$$

$$\rightarrow S_0 = F_0 e^{-cT}$$

Q: What should be included in the cost of carry (c)?

Recap: COC model

Asset classification

1. Investment assets (held to INVEST)

- Examples: gold, stock indices, currencies.
- The reality: You cannot “consume” the S&P 500 to manufacture goods.

2. Consumption assets (held to USE)

- Examples: oil, copper, wheat.
- The reality: you cannot feed a hungry herd of cattle today with a wheat futures contract.

Recap: COC model

S_0 = Spot price today

T = time until delivery date

I = PV of lump-sum income

U = PV of lump-sum storage costs

r_f = foreign riskfree rate

F_0 = Futures or forward price today

r = risk-free interest rate

q = % income yield

u = % storage costs

y = % convenience yield

Adjustments

Investment assets · equality

Held **to invest**: gold, silver, shares, indices, bonds, currencies. Holders are content to lend them out, so both arbitrages are available and the relation *binds*.

No income $F_0 = S_0 e^{rT}$

Income, PV I $F_0 = (S_0 - I) e^{rT}$

Yield q $F_0 = S_0 e^{(r-q)T}$

Currency, r_f $F_0 = S_0 e^{(r-r_f)T}$

I, q, r_f, y (-)

Consumption assets · inequality

Held **to use**: oil, copper, wheat, cattle. A refiner will not lend out the crude that keeps the plant running, so the *reverse* trade cannot be forced.

Storage, PV U $F_0 \leq (S_0 + U) e^{rT}$

Storage rate u $F_0 \leq S_0 e^{(r+u)T}$

Convenience y $F_0 = S_0 e^{(r+u-y)T}$

$y \geq 0$ is defined to close the gap

r, U, u (+)

Upper bound only!
(no equality)

One logic throughout: income earned while you carry the asset lowers F_0 , and costs of carrying it raise F_0 . The structural difference between the columns is the *direction of the arbitrage*, not the economics. Currencies sit on the left because a foreign deposit is an investment paying r_f , which is a yield by another name.

How crude oil becomes petrol

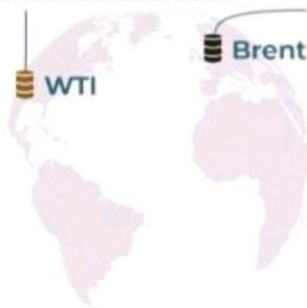
Brent and WTI are the **global benchmarks for unrefined petroleum**.
Crude oil is **heated in a furnace and distilled** into various fuels and products.

1 Extract

Crude oil is pumped out of the ground and is graded according to **thickness** and **sulfur content**



West Texas Intermediate (WTI) is sourced from US oil fields



Brent is drilled out of the North Sea between the UK and Norway

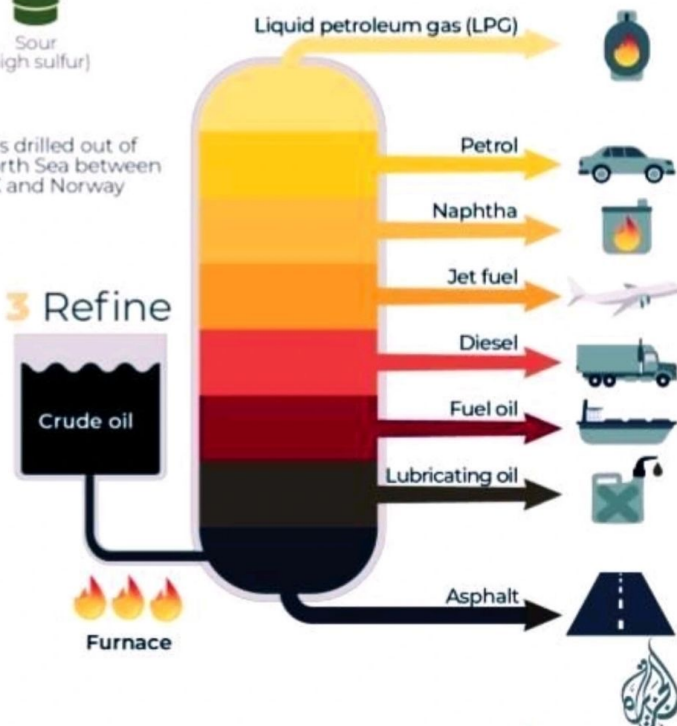
2 Transport



Oil is usually transported at sea in large oil tankers or via oil pipelines

4 Distribute

3 Refine



Option 1: Long forward today
→receive 10m barrels in 1 year

Option 2: Buy 10m barrels today
→**finance + store**→use in 1 year

Holding oil today: +interest cost
+storage cost

>>**Fair forward price reflects the net cost of carry!**



Source: OPEC Annual Statistical Bulletin 2020 | September 30, 2021

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Recap: COC model

Convenience yields (y)

- It measures the extent to which users of the commodity feel that ownership of the physical asset provides may **benefits** that are not obtained by the holders of the futures contract.
- If oil supplies are suddenly restricted, a refinery that already has physical oil in storage can keep operating.
- But these benefits are **uncertain** and **hard-to-quantify**.
- They are also firm-specific and context-specific.

Recap: COC model

- Convenience yields can **only** be incorporated into the COC model in an *ad hoc* fashion.
- It's called *ad hoc* because y is added arbitrarily to fit observed prices, not derived from first principles.

$$\begin{aligned} F_0 &\leq S_0 e^{(r+u)T} \\ \rightarrow F_0 &= S_0 e^{(r+u-y)T} \\ \rightarrow y &= r - \frac{1}{T} \ln\left(\frac{F_0}{S_0}\right) \end{aligned}$$

- For *consumption assets*, we can only obtain an **upper bound** for the futures price, but can not nail down an equality relationship between futures and spot prices.

Recap: no income

Same notation, different reference date!

Typical examples: non-dividend-paying stock; zero-coupon bonds

To generalize this example, we consider a forward contract on an investment asset with price S_0 that provides no income. Using our notation, T is the time to maturity, r is the risk-free rate, and F_0 is the forward price. The relationship between F_0 and S_0 is

$$F_0 = S_0 e^{rT} \quad (5.1)$$

If $F_0 > S_0 e^{rT}$, arbitrageurs can buy the asset and short forward contracts on the asset. If $F_0 < S_0 e^{rT}$, they can short the asset and enter into long forward contracts on it.² In our

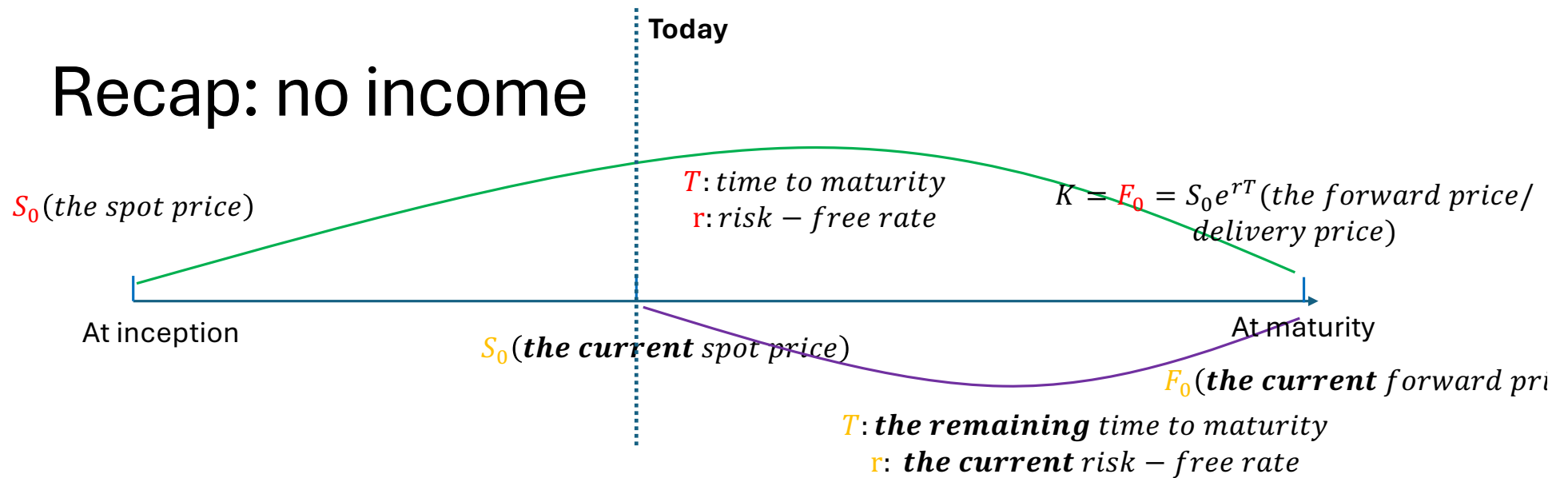
Using the notation introduced earlier, we suppose F_0 is the current forward price for a contract that was negotiated some time ago, the delivery date is T years from today, and r is the T -year risk-free interest rate. We also define:

K : Delivery price in the contract

f : Value of forward contract today

*The forward price tells us the delivery price for a new contract today, whereas f tells us what an existing contract is worth today

Recap: no income



A general result, applicable to all long forward contracts (on both investment assets and consumption assets), is

$$f = (F_0 - K)e^{-rT} \quad (5.4)$$

$$f = S_0 - Ke^{-rT} \quad (5.5)$$

Recap: no income $f = (F_0 - K)e^{-rT}$ (5.4)

The evolution of forward contract value (f)

1. At inception (Day 1):

- Entering into a forward contract normally costs nothing.
- The value of the forward contract is **zero**.
- The locked-in delivery price (K) is **set exactly equal to** the initial forward price (F_0).

$$K = F_0 \\ \rightarrow f = (0)e^{-rT} = 0$$

- Neither side has an advantage when the contract is initiated.

Q: why does K have to be exactly equal to F_0 at inception?

Recap: no income

$$F_0 = S_0 e^{rT} \quad (5.1)$$

Setup

Gold: $S_0 = 3000, r = 5\%, T = 1$, fair $F_0 = 3000 \times e^{0.05} = 3153.81$

The arbitrage

$$K > F_0$$

Forward too high: $F_0 = 3,200$

1. Borrow \$3,000, buy gold
2. Sell forward at 3,200
3. At T : deliver gold, receive 3,200; repay $3,000e^{0.05} = 3,153.81$

Riskless profit: **\$46.19**

$$K < F_0$$

Forward too low: $F_0 = 3,100$

1. Short gold, invest \$3,000 at 5%
2. Buy forward at 3,100
3. At T : investment worth 3,153.81; pay 3,100, take delivery, return the gold

Riskless profit: **\$53.81**

The left-hand trade is **cash and carry**: borrow, buy spot, carry to T , deliver. The right-hand one is **reverse cash and carry**: short the asset, lend the proceeds, take delivery. Check each against the definition: no capital committed, and no state of the world in which you lose. The next two slides do the same two trades *in symbols*, as a cash-flow ledger, and equality is the only quiet point.

Recap: no income

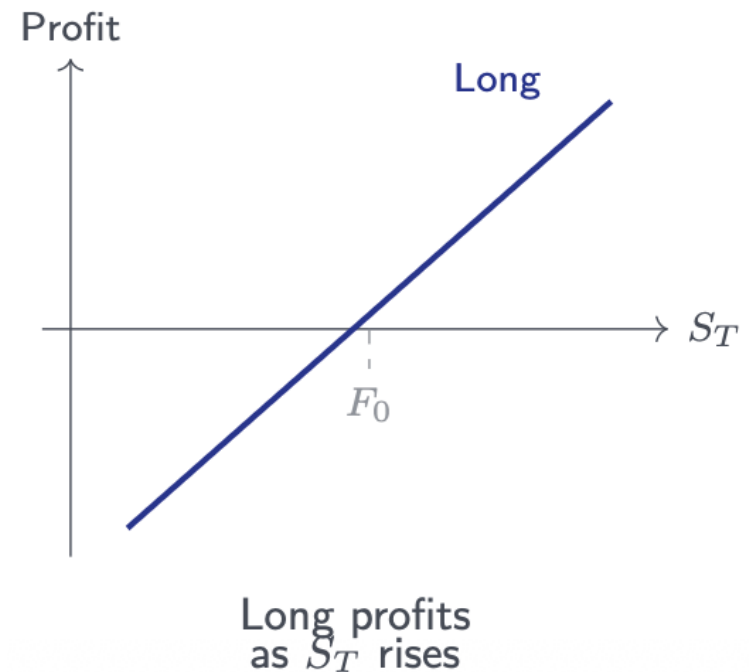
2. Over time

- The delivery price (K) remains **fixed**.
- The market's current forward price (F_0) **constantly changes**.
- Current valuation (long): $f = (F_0 - K)e^{-rT} \rightarrow \text{fluctuates}$
 - $f > 0$ (**Gain**): Market went up ($F_0 > K$). You are locked in at a cheaper price.
 - $f < 0$ (**Loss**): Market went down ($F_0 < K$). You are stuck paying a higher price.
- For the value of a **short** forward contract: $f = -(S_0 - Ke^{-rT})$

Recap: no income

3. At maturity

- At maturity, $T = 0$
- $\rightarrow f_T = (F_T - K)e^{-rT} = F_T - K$
- Using $F_T = S_T$, and $K = F_0$
- $\rightarrow f_T = S_T - F_0$
- **The forward contract value becomes its payoff!**



Problem 5.9

- *A one-year long forward contract on a non-dividend-paying stock is entered into when the stock price is \$40 and the risk-free rate of interest is 10% per annum with continuous compounding.*
 - (a) *What are the forward price and the initial value of the forward contract?*
 - (b) *Six months later, the price of the stock is \$45 and the risk-free interest rate is still 10%. What are the forward price and the value of the forward contract?*

Answer 5.9

Setup

- Strategy: Long forward contract
- Asset: non-dividend-paying stock (no income)
- Initial spot price $S_0 = 40$
- Risk-free rate: $r = 10\%$ (*continuous compounding*)
- Time to maturity: $T = 1$
- $F_0 = ?$; $f_0 = ?$
- *Current $S_0 = 45$; Current $T = \frac{1}{2}$; Current $r = 10\%$*
- *Current $F_0 = ?$; $f = ?$*

Answer 5.9

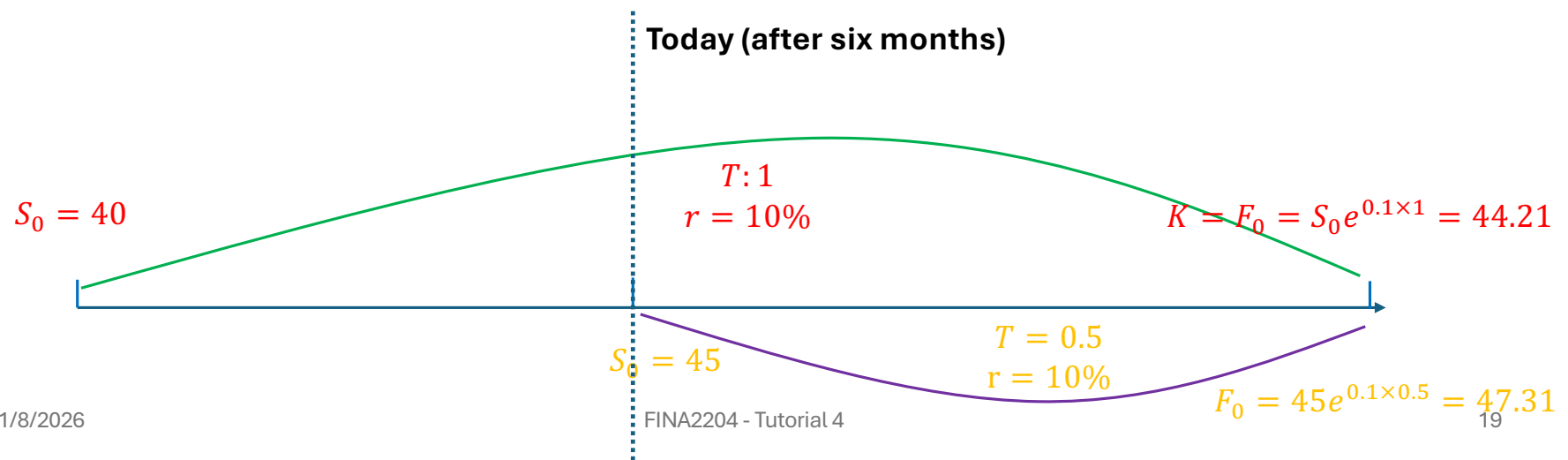
$$F_0 = S_0 e^{rT} \quad (5.1)$$

a) The forward price, F_0 , is given by **equation (5.1)** as:

$$F_0 = 40e^{0.1 \times 1} = 44.21$$

Or \$44.21.

The initial value of the forward contract is **zero**.



Answer 5.9

$$f = (F_0 - K)e^{-rT} \quad (5.4)$$

b) The current forward price is:

$$45e^{0.1 \times 0.5} = 47.31$$

Or \$47.31.

The delivery price K in the contract is \$44.21. The value of the contract, f , after six months is given by **equation (5.4)** as:

$$f = (47.31 - 44.21)e^{-0.1 \times 0.5} = 2.95$$

i.e., it is \$2.95.

Recap: income, PV /

Typical examples: dividend-paying stocks; coupon-bearing bonds

We can generalize from this example to argue that, when an investment asset provides income with a present value of I during the life of a forward contract, we have

$$F_0 = (S_0 - I)e^{rT} \quad (5.2)$$

I : PV of lump-sum income

Similarly, using equation (5.4) in conjunction with (5.2) gives the following expression for the value of a long forward contract on an investment asset that provides a known income with present value I :

$$f = S_0 - I - Ke^{-rT} \quad (5.6)$$

proof: $f = ((S_0 - I)e^{rT} - K)e^{-rT} = S_0 \underbrace{e^{rT}e^{-rT}}_{=1} - I \underbrace{e^{rT}e^{-rT}}_{=1} - Ke^{-rT} = S_0 - I - Ke^{-rT}$

Recap: yield q

Typical assets: stock market indices; foreign currencies

Define q as the average yield per annum on an asset during the life of a forward contract with continuous compounding. It can be shown (see Problem 5.20) that

$$F_0 = S_0 e^{(r-q)T} \quad (5.3)$$

q : % income yield

Finally, using equation (5.4) in conjunction with (5.3) gives the following expression for the value of a long forward contract on an investment asset that provides a known yield at rate q :

$$f = S_0 e^{-qT} - K e^{-rT} \quad (5.7)$$

$$\text{proof: } f = (S_0 e^{(r-q)T} - K) e^{-rT} = S_0 e^{-qT} \underbrace{e^{rT} e^{-rT}}_{=1} - K e^{-rT} = S_0 e^{-qT} - K e^{-rT}$$

Problem 5.4

- *A stock index currently stands at 350. The risk-free rate is 8% per annum (with continuous compounding) and the dividend yield on the index is 4% per annum.*
- *What should the futures price for a four-month contract be?*

Answer 5.4

Setup

- Asset: stock index (yield q)
- Initial spot price: $S_0 = 350$
- Risk-free rate: $r = 8\%$
- Dividend yield: $q = 4\%$
- Time to maturity: $T = 4/12 \approx 0.3333$
- $F_0 = ?$

Answer 5.4

$$F_0 = S_0 e^{(r-q)T} \quad (5.3)$$

- The future price is

$$F_0 = 350 \times e^{(0.08-0.04) \times 0.3333} = 354.7$$

Or \$354.7.

Problem 5.10

- *The risk-free rate of interest is 7% per annum with continuous compounding, and the dividend yield on a stock index is 3.2% per annum. The current value of the index is 150.*
- *What is the six-month futures price?*

Answer 5.10

Setup

- Asset: stock index
- Initial spot price: $S_0 = 150$
- Risk-free rate: $r = 7\%$
- Dividend yield: $q = 3.2\%$
- Time to maturity: $T = \frac{6}{12} = 0.5$
- $F_0 = ?$

Answer 5.10 $F_0 = S_0 e^{(r-q)T}$ (5.3)

Using equation (5.3) the six month futures price is
 $150e^{(0.07-0.032)\times 0.5} = 152.88$

Or \$152.88.

Problem 5.12

- *Suppose that the risk-free interest rate is 10% per annum with continuous compounding and that the dividend yield on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405.*
- *What arbitrage opportunities does this create?*

Answer 5.12

Setup

- Asset: stock index
- Initial spot price: $S_0 = 400$
- Risk-free rate: $r = 10\%$
- Dividend yield: $q = 4\%$
- Time to maturity: $T = 4/12$
- *Actual* $F_0 = 405$
- *fair* $F_0 = ?$

Answer 5.12 $F_0 = S_0 e^{(r-q)T}$ (5.3)

The theoretical futures price is

$$400e^{(0.10-0.04) \times 4/12} = 408.08$$

The actual futures price is only 405. This shows that the index futures price is too low relative to the index.

The correct arbitrage strategy (**reverse cash & carry**) is

a) buy futures contracts

b) short the shares underlying the index.

• **This strategy yields a risk-free arbitrage profit of \$3.08.**

Answer 5.12

Known income yield q : reverse cash and carry

If $F_0 < S_0 e^{(r-q)T}$:

	Strategy	Cash flow	
<i>Time 0</i>	Buy futures @ F_0	0	
	Sell short $S_0 e^{-qT}$ of stock	$+S_0 e^{-qT}$	$\rightarrow 400e^{-0.04 \times 4/12} = +394.70$
	Lend $S_0 e^{-qT}$ till T	$-S_0 e^{-qT}$	$\rightarrow -394.70$
		0	
<i>Time T</i>	Sell futures @ F_T	$+F_T - F_0$	$\rightarrow -405$
	Receive loan plus interest	$+S_0 e^{(r-q)T}$	$\rightarrow 400e^{(0.10-0.04) \times 4/12} = 408.08$
	Buy stock and return it	$-S_T$	
		$-F_0 + S_0 e^{(r-q)T} > 0$	

$$\rightarrow -405 + 408.08 = +3.08$$

Problem 5.19

- *Show that the growth rate in an index futures price equals the excess return of the portfolio underlying the index over the risk-free rate. Assume that the risk-free interest rate and the dividend yield are constant.*

Answer 5.19

- Define the portfolio's excess return over the risk-free rate as x .
- **Total return** equals the risk-free rate (r) plus the excess return for taking market risk (x).

$$r + x$$

- **Capital gains:** since a portion of the total return ($r + x$) is paid out as cash dividends (q), the price itself grows at a slower rate.

$$r + x - q$$

- Knowing this, the future spot price S_1 at t_1 can be expressed as:

$$S_1 = S_0 e^{(r+x-q)t_1}$$

Answer 5.19

$$F_0 = S_0 e^{(r-q)T} \quad (5.3)$$

- Suppose that F_0 is the futures price at time zero for a contract maturing at time T and F_1 is the futures price for the same contract at time t_1 . It follows that

$$F_0 = S_0 e^{(r-q)T}; \quad F_1 = S_1 e^{(r-q)(T-t_1)}$$

These equations imply that

$$\frac{F_1}{F_0} = \frac{S_1}{S_0} e^{-(r-q)t_1} = \frac{S_0 e^{(r+x-q)t_1}}{S_0} e^{-(r-q)t_1} = e^{xt_1}$$

The futures price grows at rate x

Recap: Currency, r_f

A foreign currency has the property that the holder of the currency can earn interest at the risk-free interest rate prevailing in the foreign country. For example, the holder can invest the currency in a foreign-denominated bond. We define r_f as the value of the foreign risk-free interest rate when money is invested for time T . The variable r is the domestic risk-free rate when money is invested for this period of time.

The relationship between F_0 and S_0 is

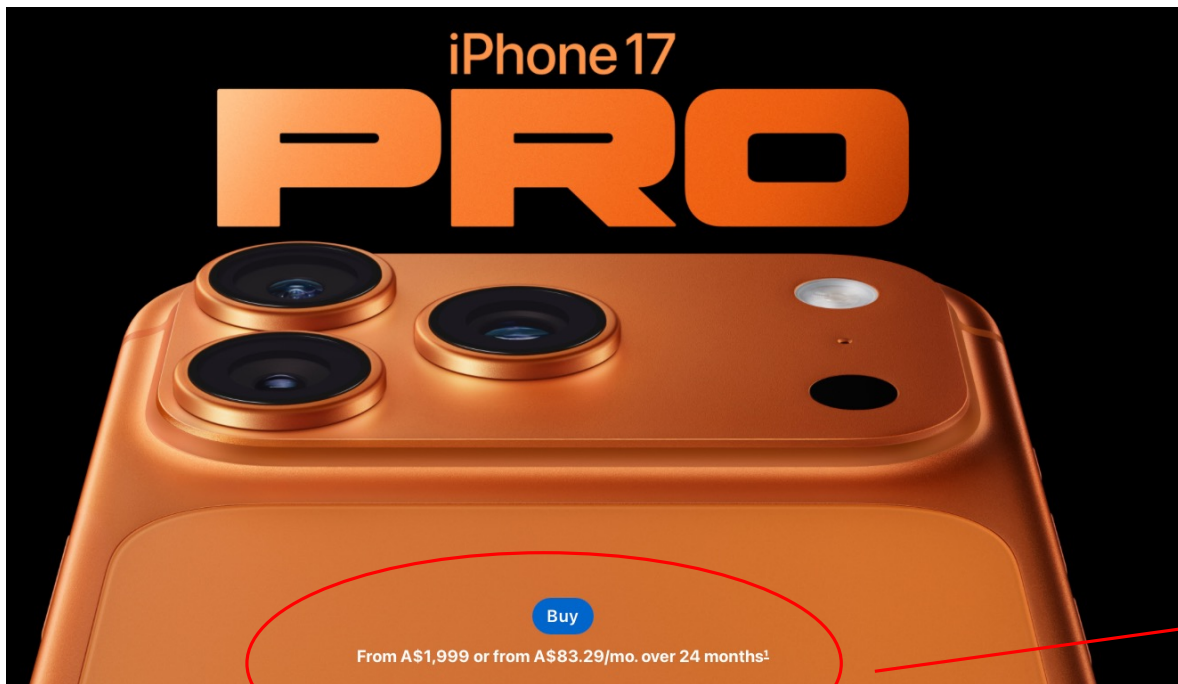
$$F_0 = S_0 e^{(r-r_f)T} \quad (5.9)$$

Note that equation (5.9) is identical to equation (5.3) with q replaced by r_f . This is not a coincidence. A foreign currency can be regarded as an investment asset paying a known yield. The yield is the risk-free rate of interest in the foreign currency.

$$F_0 = S_0 e^{(r-q)T} \quad (5.3)$$

Recap: Currency, r_f

Left is foreign; right is domestic!



foreign asset (currency) domestic currency

Exchange-rate quote:
USD1 = AUD1.556



Price tag:
1 iPhone = AUD 1,999

foreign asset domestic currency

Problem 5.14

- *The two-month interest rates in Switzerland and the United States are 2% and 5% per annum, respectively, with continuous compounding. The spot price of the Swiss franc is \$0.8000. The futures price for a contract deliverable in two months is \$0.8100.*
- *What arbitrage opportunities does this create?*

Answer 5.14

Setup

foreign asset (currency)

domestic currency


$$CHF1 = USD0.8000$$

- Domestic risk-free rate: $r = 5\%$
- Foreign risk-free rate: $r_f = 2\%$
- Spot price: $S_0 = 0.8000$
- Time to maturity: $T = 2/12$
- Actual futures price: $F_0 = 0.8100$
- Fair future price: $F_0 = ?$

Answer 5.14

$$F_0 = S_0 e^{(r-r_f)T} \quad (5.9)$$

The theoretical futures price is

$$0.8000 e^{(0.05-0.02) \times 2/12} = 0.8040$$
$$\text{fair } F_0 = 0.8040 < \text{actual } F_0 = 0.8100$$

- The actual futures price is too high.
- This suggests that an arbitrageur should buy Swiss francs and short Swiss francs futures (**cash & carry**).

Problem 5.17

- *When a known future cash outflow in a foreign currency is hedged by a company using a forward contract, there is no foreign exchange risk. When it is hedged using futures contracts, the daily settlement process does leave the company exposed to some risk. Explain the nature of this risk. In particular, consider whether the company is better off using a futures contract or a forward contract when*

Problem 5.17

- a) *The value of the foreign currency **falls** rapidly during the life of the contract*
- b) *The value of the foreign currency **rises** rapidly during the life of the contract*
- c) *The value of the foreign currency **first rises and then falls back** to its initial value*
- d) *The value of the foreign currency **first falls and then rises back** to its initial value*

Answer 5.17

Setup

- The position: To hedge a future cash outflow, the company must buy the foreign currency. Therefore it takes **a long position**.
- For a long position: price goes up $\rightarrow$ gain; price goes down $\rightarrow$ loss.

Forward vs. futures (when the gains/losses are realized)

1. Forward contract: the value of f is just an *unrealized paper profit/loss*. No cash changes hands until the final day.
2. Futures contract: the value f is realized every day due to mark-to-market.

Answer 5.17

Setup (cont.)

- If $f > 0$, you receive cash early and earn interest income.
- If $f < 0$, you must pay margin calls (borrow cash) and suffer interest costs.
- The core risk is the **timing of cash flows** and the resulting interest costs/gains.

Rule

Positive cash flow early → futures is better.

Negative cash flow early → forward is better.

Answer 5.17

Case A: the foreign currency falls rapidly

a) In this case, the **forward contract** would lead to a slightly better outcome.

The company will make a loss on its hedge ($f < 0$).

If the hedge is with a forward contract, the whole of the loss will be realized at the end.

If it is with a futures contract, the loss will be realized day by day throughout the contract. On a present value basis the former is preferable.

Answer 5.17

Case B: the foreign currency rises rapidly.

b) In this case, the **futures contract** would lead to a slightly better outcome. The company will make a gain on the hedge ($f < 0$). If the hedge is with a forward contract, the gain will be realized at the end. If it is with a futures contract, the gain will be realized day by day throughout the life of the contract. On a present value basis the latter is preferable.

Answer 5.17

Case C: the currency rises first and falls later.

c) In this case, the **futures contract** would lead to a slightly better outcome. This is because it would involve positive cash flows early and negative cash flows later.

Case D: the currency falls first and rises later.

d) In this case, the **forward contract** would lead to a slightly better outcome. This is because, in the case of the futures contract, the early cash flows would be negative and the later cash flow would be positive.

Recap: Storage, PV U

PV!

Storage costs can be treated as negative income. If U is the present value of all the storage costs during the life of a forward contract, it follows from equation (5.2) that

$$F_0 = (S_0 + U)e^{rT} \quad (5.11)$$

Example 5.8 provides an application of this formula. Assume convenience yield is zero ($y=0$)

If the storage costs (net of income) incurred at any time are proportional to the price of the commodity, they can be treated as negative yield. In this case, from equation (5.3), we have

$$F_0 = S_0 e^{(r+u)T} \quad (5.12)$$

where u denotes the storage costs per annum as a proportion of the spot price net of any yield earned on the asset.

Problem 5.15

- *The current price of silver is \$30 per ounce. The storage costs are \$0.48 per ounce per year payable quarterly in advance. Assuming that interest rates are 10% per annum for all maturities, calculate the futures price of silver for delivery in nine months.*

Answer 5.15

$$F_0 = (S_0 + U)e^{rT}$$

(5.11)

The present value of the storage costs for nine months are

$$U = 0.12 + 0.12e^{-0.10 \times 0.25} + 0.12e^{-0.10 \times 0.5} = 0.351$$

Or \$0.351.

The futures price is from **equation (5.11)** given by F_0 where

$$F_0 = (30 + 0.351)e^{0.10 \times 0.75} = 32.72$$

i.e., it is \$32.72 per ounce.

